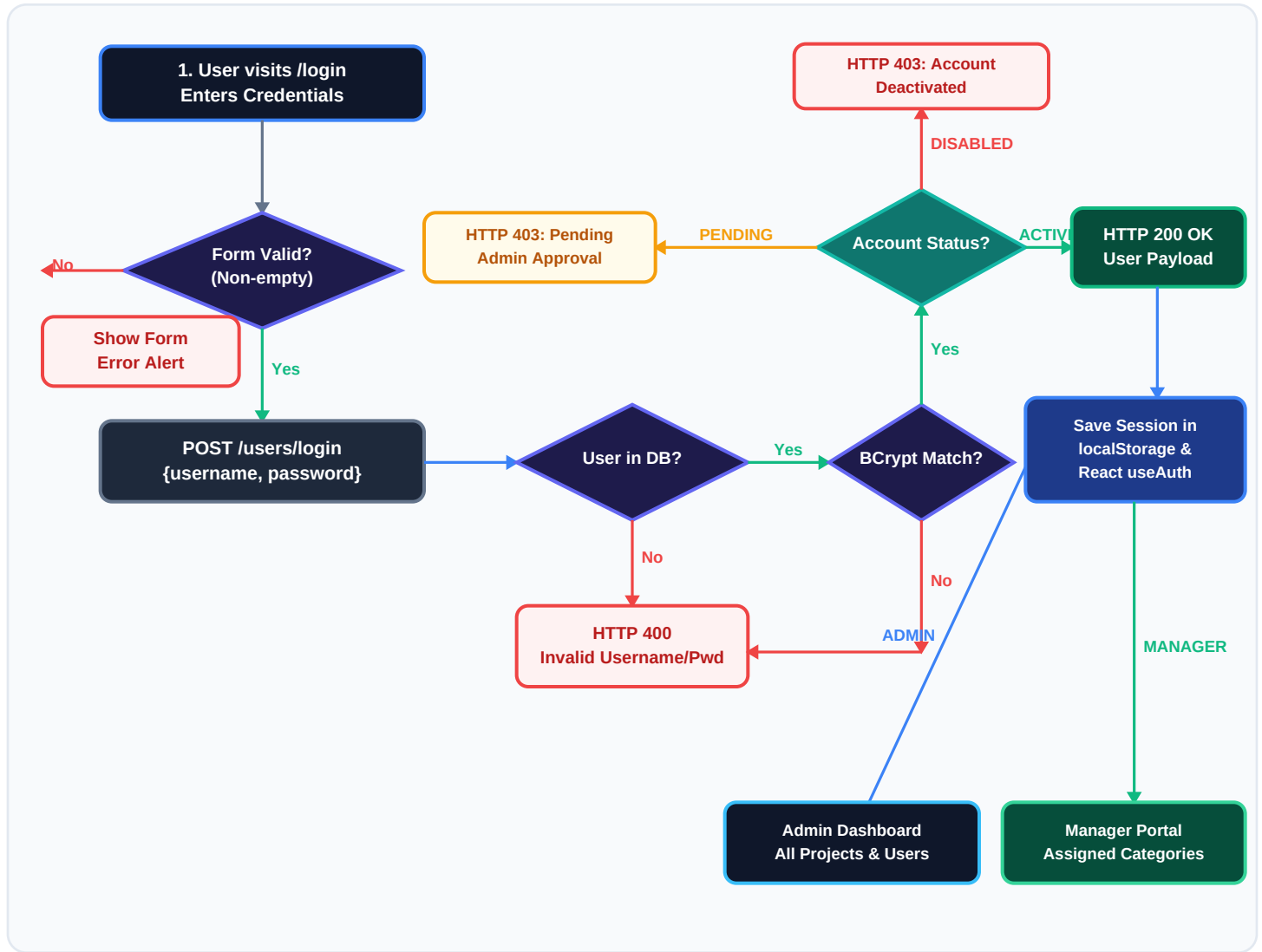


User Sign In & Authentication Flow

Technical Flowchart & Architectural Verification Guide



Authentication Logic & Security Rules

Phase	Component	Condition / Rule	Outcome
Client Check	LoginPage.jsx	Username/Password non-empty	Sends POST request or displays inline validation error.
User Lookup	UserService.java	<code>userRepo.findByUsername(name)</code>	Validates username existence; returns HTTP 400 if not found.
Password Verification	BCryptPasswordEncoder	<code>passwordEncoder.matches(raw, hash)</code>	Salted hash verification; returns HTTP 400 if invalid.
Status: Pending	UserService.java	<code>status == 'PENDING_APPROVAL'</code>	Blocks login with HTTP 403 until approved by Admin.
Status: Disabled	UserService.java	<code>status == 'DISABLED'</code>	Blocks login with HTTP 403; account is deactivated.
Session Storage	<code>useAuth.js</code> / <code>request.js</code>	<code>status == 'ACTIVE'</code>	Stores user payload in localStorage; attaches X-User-Id / X-Role.
Role Navigation	ProtectedRoute.jsx	ADMIN / MANAGER / SUPERVISOR	Routes to scoped dashboard with assigned categories & projects.